

Example 1-8: Repeat Examples 1-5 and 1-6 using 9's complements.

(a) $M = 72532$
 $N = 03250$
 9's complement of $N = \begin{array}{r} 99999 \\ -03250 \\ \hline 96749 \end{array}$

$$\begin{array}{r} 72532 \\ + 96749 \\ \hline 169281 \end{array}$$

1 → end-around carry
 $\begin{array}{r} 69281 \\ + 1 \\ \hline 69282 \end{array}$

(b) $M = 03250$
 $N = 72532$
 9's complement of $N = \begin{array}{r} 99999 \\ -72532 \\ \hline 27467 \end{array}$

$$\begin{array}{r} 03250 \\ + 27467 \\ \hline 30717 \end{array}$$

no carry

9's complement of 30717: $\begin{array}{r} 99999 \\ -30717 \\ \hline 69282 \end{array}$

answer: $-69282 \square$

Example 1-9: Repeat Example 1-7 using 1's complement.

(a) $M = 1010100$
 $N = 1000100$
 1's complement of $N = 0111011$

$$\begin{array}{r} 1010100 \\ + 0111011 \\ \hline 10001111 \end{array}$$

1 → end-around carry
 $\begin{array}{r} 0001111 \\ + 1 \\ \hline 0010000 \end{array}$

answer: 10000

(b) $M = 1000100$
 $N = 1010100$
 1's complement of $N = 0101011$

$$\begin{array}{r} 1000100 \\ + 0101011 \\ \hline 1101111 \end{array}$$

→ no carry.

1's complement of 1101111 = 0010000
 answer: -10000

Problems

1-1. Write the first 20 decimal digits in base 3.

$0: 0 \times 3^0 = 000$
 $1: 1 \times 3^0 \rightarrow 001$
 $2: 2 \times 3^0 \rightarrow 002$
 $3: 1 \times 3^1 + 0 \times 3^0 \rightarrow 010$
 $4: 1 \times 3^1 + 1 \times 3^0 \rightarrow 011$